

# A Web-Based VR-Testbed for Situated Urban Heat Visualization: GIS Data Pipeline, Visualization Variants, and Design Implications for Mobile AR

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**Abstract:** The communication of invisible urban environmental hazards such as heat stress to a lay audience continues to represent a central challenge in urban risk communication. This paper presents a web-based VR prototype that integrates high-resolution urban heat map data ( $10 \times 10$  m, Berlin Climate Analysis 2022) and real-time weather data from an API into a 3D urban environment derived from OSM using Three.js. The prototype serves as a controlled test environment for evaluating situational visualization designs prior to implementation in mobile AR. We systematically compare two geometric representations (heatmap vs. heightmap), three color scales (Vik, Magma, Jet), two normalization strategies (global vs. local range), as well as Gaussian interpolation. A qualitative visual evaluation from the first-person perspective (FPP) shows that flat heatmaps in combination with the perceptually uniform, diverging Vik color scale and local-range normalization provide the best spatial readability at street level, whereas heightmaps suffer from occlusions and impair the readability of the environment. We derive concrete design principles for future mobile AR applications and identify key trade-offs between comparability, perception of detail, and data accuracy. Limitations include the absence of a formal user study as well as scalability constraints of the current prototype.

**Keywords:** Urban Heat Visualization, Situated Visualization, Web-based VR, Mobile Augmented Reality, GIS Data Pipeline

## 1 Introduction

Cities are central living environments of modern society and, at the same time, are particularly affected by the impacts of climate change. High population densities, sealed surfaces, and urban infrastructure promote the formation of urban heat islands, causing cities to warm more strongly than their surrounding areas (cf. [Ab25]; [Un24]). Especially during heat waves, this increases the health burden on the population and can lead to heat stress, circulatory problems, and increased mortality (cf. [Da24]; [Wi23]; [Eb21]). Although extensive environmental and geospatial data are available, the associated risks remain difficult for many citizens to understand. One reason is the often abstract representation through maps, diagrams, or reports, which establish only a limited connection to the real

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environment (cf. [Sh12]; [HB21]). In particular, traditional two-dimensional visualizations make the intuitive perception of spatial environmental phenomena more difficult (cf. [MK01]; [Sh12]).

An alternative approach is provided by the concept of *Situated Visualization*, in which digital information is displayed directly within its spatial context (cf. [Wh09]). Technologies such as Augmented Reality (AR) and Virtual Reality (VR) enable such context-based visualizations and can support the understanding of complex environmental information. Within the scope of this work, a prototype for the visualization of urban heat data in a virtual city environment was developed. VR provides a suitable test environment for immersively presenting and efficiently evaluating visualization concepts (cf. [JS25]; [LVF22]). At the same time, VR can be used as a simulation environment for AR applications, thereby reducing development effort and hardware requirements (cf. [LVF22]). However, limitations remain regarding the transferability to real urban environments (cf. [Tr23]).

Despite the growing body of research on immersive environmental visualizations, no reproducible web-based VR testbed currently exists that systematically compares urban heat visualization variants from the first-person perspective of a pedestrian. In particular, the effects of geometry type, color scale, normalization strategy, and spatial interpolation on the perception of microclimatic heat patterns remain largely unexplored. Against this background, the present work develops a reproducible GIS data pipeline for processing urban heat data and implements an open WebVR prototype as a test environment for AR-related visualizations. Based on this, different representations of microclimatic data are compared, and design implications for future AR applications are derived.

The remainder of this paper is structured as follows. Section 2 reviews the foundations of urban heat visualization, situated visualization, and immersive technologies, and discusses related work. Section 3 presents the system architecture, including the GIS data pipeline, data sources, and WebVR implementation. Section 4 describes the investigated visualization designs and variants. Section 5 reports the qualitative evaluation and compares the different visualization approaches. Section 6 discusses the results, derives design implications for mobile AR applications, and outlines the limitations of the current work. Finally, Section 7 summarizes the main contributions and highlights directions for future research.

## 2 Background & Related Work

Cities are particularly affected by heat stress due to high building densities, sealed surfaces, and anthropogenic heat sources. The primary cause is the *urban heat island effect*, in which urban areas store heat and cool down more slowly than their surrounding regions (cf. [Ok73]). Especially during heat waves, this can lead to health impacts such as heat stress, circulatory problems, and increased mortality (cf. [Wi23]; [Eb21]). Against the backdrop of climate change, the analysis and communication of such urban environmental and climate

risks are becoming increasingly important for urban planning and climate adaptation (cf. [WBR13]).

At the same time, extensive environmental and geospatial datasets on temperature, air quality, and precipitation are available today (cf. [Ar25]). Nevertheless, challenges remain regarding their communication. Traditional forms of representation such as maps, diagrams, or reports often require domain-specific knowledge and make it difficult, particularly for non-experts, to interpret spatial relationships (cf. [Wa19]; [Sh12]). In addition, many representations lack a direct connection to the real environment, making location-dependent phenomena such as heat exposure only partially understandable (cf. [Sh12]; [Wh09]).

One approach to addressing these challenges is the concept of *Situated Visualization*. It describes the presentation of digital information directly within its spatial context of origin, allowing data to be interpreted in conjunction with the real environment (cf. [Wh09]). Augmented Reality (AR) supports this approach by embedding virtual information directly into the physical environment (cf. [Az97]; [Wh09]). As a result, environmental and health risks can be perceived more intuitively, and spatial relationships can be understood more easily (cf. [Sh12]; [Va24]).

Initial research projects have already employed immersive technologies to visualize urban environmental and climate data. The *Urban Climate InteracTable* project visualizes heat data within a mixed reality environment (cf. [Re25]), while *GLARA* uses AR and VR technologies to communicate urban heat in an understandable way (cf. [GL23]). These projects demonstrate the potential of immersive technologies for communicating urban climate phenomena. However, they primarily focus on application-specific implementations and user-facing experiences. In contrast, the present work focuses on the development of a reproducible, open-source WebVR testbed that enables a systematic comparison of urban heat visualization variants.

Virtual prototypes have proven to be an established method for the development and evaluation of such concepts (cf. [JS25]). In particular, Virtual Reality (VR) provides an immersive test environment in which visualization concepts can be efficiently implemented, examined, and iteratively refined (cf. [LVF22]). Furthermore, VR can serve as a simulation environment for AR applications, enabling visualization concepts to be analyzed under controlled conditions without relying on real-world test environments (cf. [LVF22]). However, limitations remain regarding visual quality and the complete representation of real urban contexts (cf. [Tr23]).

### 3 System Architecture

The developed application consists of four central components: the data sources, a Python-based processing pipeline, standardized export formats, and the WebVR application. Figure 1 provides an overview of the system architecture.

Geospatial data of the urban environment and raster-based heat data serve as input data. These datasets are imported, cleaned, and prepared for visualization within a Python pipeline. Subsequently, the processed data are exported in JSON and CSV formats. The visualization is performed in a web-based VR application built on Three.js. When the application is launched, the exported datasets are loaded and displayed as interactive 3D objects within the virtual urban environment.

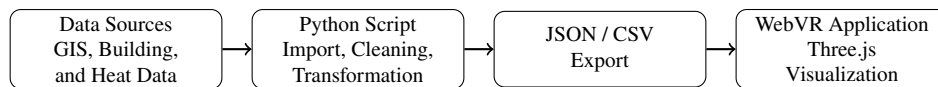


Fig. 1: Overview of the system architecture. Data sources are processed within a Python pipeline and provided as JSON and CSV files for the Three.js-based WebVR application.

### 3.1 Data Sources

Datasets from three different sources were used for the visualization of heat exposure. First, the city-wide climate simulations (*Klimaanalyse*) of the State of Berlin, which have been conducted regularly since 1991<sup>2</sup>, serve as the primary data source. These datasets are freely available through the *Berlin Open Data* portal under the *Deutschland Zero* data license (Version 2.0)<sup>3</sup> as georeferenced raster data in GeoTIFF format<sup>4</sup>. The climate simulations are based on the *FITNAH 3D* flow model, which simulates urban heat islands while considering buildings, vegetation, and surface sealing (cf. [Gr92]; [Gr02]; [Gr14]). In addition, current weather data were integrated via the API of the *Open-Meteo* service. These provide dynamic real-time temperatures based on meteorological models and enable the visualization to be updated with current weather information. For spatial contextualization, geospatial data from *OpenStreetMap* (OSM) were used. These form the basis for generating the three-dimensional city model and enable the embedding of temperature information into a realistic urban environment (cf. [Bi15]). Satellite-based remote sensing data were deliberately excluded because they primarily capture surface temperatures and therefore allow only limited conclusions regarding air temperature, which is more relevant to human thermal exposure (cf. [Zh25]).

### 3.2 Data Selection and Processing

The available raster datasets from the Berlin Climate Analysis were evaluated according to the quality criteria defined in DIN EN ISO 19157. Particular consideration was given to completeness, consistency, positional accuracy, temporal quality, and thematic accuracy. This analysis led to the exclusion of older datasets from 2001 and 2005, as they exhibit

<sup>2</sup> cf. <https://www.berlin.de/umweltatlas/klima/klimaanalyse/>

<sup>3</sup> cf. <https://www.govdata.de/dl-de/zero-2-0>

<sup>4</sup> cf. [https://daten.berlin.de/datensaetze?q=klimamodell&sort=score+desc%2C+metadata\\_modified+desc](https://daten.berlin.de/datensaetze?q=klimamodell&sort=score+desc%2C+metadata_modified+desc)

methodological differences as well as limitations regarding comparability and data structure. Instead, the 2022 Climate Analysis was selected as a consistent and high-resolution data source. The temperature data are used to represent current heat exposure, while long-term developments are visualized using the projected heat-day indicators provided by the climate analysis.

As an result an example data set was generated by this pipline, that was used for further investigation of the visualisation in the prototype.

## 4 Visualization Design and Variants

The visualization of urban temperature data in immersive environments requires several design decisions that influence perceptual accuracy, spatial understanding, and the interpretation of thermal patterns. To systematically investigate this design space, a morphological box that decomposes the visualisation into four dimensions was defined: *geometric encoding*, *color encoding*, *normalization strategy*, and *interpolation approach* (figure 2). A morphological box is a Matrix, that shows multiple independent dimensions, which each can have different characteristics [Ri98].

|                    |          |       |                        |  |
|--------------------|----------|-------|------------------------|--|
| Geometric encoding | Heatmap  |       | Heightmap              |  |
| Color encoding     | Vik      | Magma | Jet                    |  |
| Normalization      | Local    |       | Global                 |  |
| Interpolation      | Raw Data |       | Gaussian Interpolation |  |

Fig. 2: Morphological box of the investigated visualization design space.

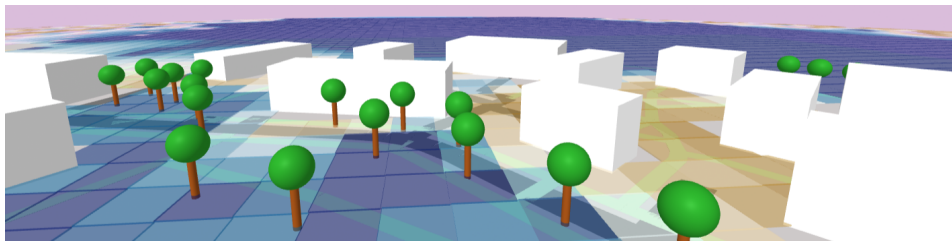


Fig. 3: Bird's-eye view of the VR environment showing a heatmap visualization using the Vik colormap with local normalization (temperature range: 20.8–34 °C).

The selected dimensions represent core visualization parameters in geospatial and scientific visualization. Geometric encoding defines how scalar temperature values are mapped into spatial form and thereby affects depth perception and occlusion, consistent with established approaches for terrain and 3D scalar field visualization [TL10]. Color encoding determines how temperature values are translated into perceptual differences in hue and luminance, directly influencing the detectability of gradients and hotspots, and is grounded in well-studied principles of perceptually motivated colormap design [BT07; CSH20]. Normalization defines how raw temperature values are mapped to a fixed color scale range, thereby controlling whether absolute comparability or local contrast is emphasized, reflecting known effects of value-to-color mapping on perceptual comparison tasks [Wa19]. Finally, interpolation governs whether the data are rendered as discrete observations or as a spatially continuous field, shaping the perceived smoothness of temperature distributions and following standard practices in geospatial surface reconstruction [BML15]. The following sections systematically detail each dimension of the morphological box and its corresponding characteristics.

#### 4.1 Geometric encoding

Two geometric encodings were evaluated: a planar heatmap and a three-dimensional heightmap. The heatmap projects temperature values onto a planar surface using color as the sole visual variable. This representation is expected to preserve the visibility of contextual urban structures and minimizes occlusion, which is particularly beneficial in dense urban environments and situated visualization scenarios [AN+25; HNS25; SCS23]. In contrast, the heightmap encodes temperature values as vertical displacement, forming a three-dimensional scalar field. This additional geometric dimension can improve the perception of spatial structures and facilitate the identification of local extrema [TL10]. However, it also introduces occlusion effects and can reduce the simultaneous visibility of spatial regions, potentially affecting overall spatial interpretability.

#### 4.2 Color encoding

Color encoding constitutes the primary visual variable for representing temperature and therefore strongly influences pattern recognition and data interpretation. Three color scales were investigated: *Jet*, *Magma*, and *Vik*. *Jet* was included as a reference due to its widespread use in legacy visualization systems. However, previous studies have shown that its non-uniform luminance distribution may create artificial boundaries and distort the perception of continuous data [BT07]. In contrast, perceptually uniform color scales preserve proportional relationships between numerical and perceived differences, thereby supporting more accurate interpretation [Cr18]. *Magma* represents a sequential color scale optimized for encoding absolute magnitudes, whereas *Vik* is a diverging scale designed to emphasize deviations relative to a central reference value. Since urban heat analyses involve

both the identification of hotspots and the assessment of local temperature anomalies, both sequential and diverging encodings were considered relevant.

### **4.3 Normalisation of the color scale**

Temperature values must be normalized prior to mapping them to a fixed color scale. Two normalization strategies were used: global normalization and local normalization. Global normalization applies a fixed temperature range across all visualizations. In this work, the global range is derived from an estimated urban temperature interval of 10°C to 40°C. This approach ensures that identical colors correspond to identical absolute temperature values across all scenes, enabling direct comparison between different spatial regions and datasets. However, if the actual observed range is narrower, parts of the color scale may remain unused, potentially reducing local contrast sensitivity. Local normalization computes the minimum and maximum temperature values within the currently visualized dataset or spatial subset. This approach maximizes the utilization of the color scale for each individual view, thereby enhancing local contrast and pattern visibility. However, it sacrifices direct comparability of absolute temperature values across different scenes, as identical colors may correspond to different numerical values.

### **4.4 Interpolation**

Interpolation techniques are commonly used in geospatial visualization to derive continuous scalar fields from discrete measurements [BML15]. Two approaches were considered in this work: raw data rendering and Gaussian interpolation. The raw data representation directly visualizes the original grid structure without additional smoothing. This preserves measurement fidelity and avoids introducing artificial values, but may result in visually discontinuous surfaces. Gaussian interpolation generates a smoothed continuous scalar field using a grid-based Gaussian kernel interpolation with a 1-unit resolution grid and a kernel parameter of  $\sigma = 10$ . The method incorporates a distance-based truncation strategy to limit extrapolation effects. The resulting field is normalized within the interpolation process. This approach improves visual continuity and supports the perception of large-scale spatial gradients, but may attenuate sharp local variations.

### **4.5 Implementation of Visualization Variants**

To demonstrate the characteristics of the morphological box, the identified dimensions were implemented in a VR scene using an example dataset. Based on this design space, multiple visualization variants were generated and compared within the prototype, as shown in table 1. The full factorial combination of all four dimensions was not exhaustively evaluated.

Tab. 1: Overview of the implemented visualization variants derived from the morphological box and evaluated within the VR prototype.

| Variant            | 1       | 2       | 3         | 4       | 5       | 7          | 8        | 9        |
|--------------------|---------|---------|-----------|---------|---------|------------|----------|----------|
| Geometric encoding | heatmap | heatmap | heightmap | heatmap | heatmap | heightmap  | heatmap  | heatmap  |
| Color encoding     | Vik     | Vik     | Vik       | Jet     | Magma   | Vik        | Vik      | Magma    |
| Normalisation      | Local   | Global  | Local     | Local   | Local   | Local      | Local    | Local    |
| Interpolation      | None    | None    | None      | None    | None    | None       | Gaussian | Gaussian |
| Figure             | 3, 6c   | 4       | 5         | 6a      | 6b      | 7a, 7b, 7c | 8a       | 8b       |

Instead, the visualization design space was explored iteratively in a stepwise process. Initial comparisons focused on normalization strategies using a heatmap representation. Figure 3 shows the local characteristic with 20.8–34 °C and figure 4 the global one. Subsequently, heatmap (figure 3) and heightmap (figure 5) representations were compared under a fixed local normalization and consistent color scheme (Vik).

In a further step, different color scales (Jet, Magma, Vik) were evaluated using the heatmap (figure 6) representation under local normalization. The heightmap representation was then explored primarily using the Vik color scale with varying extrusion heights (figure 7). Finally, Gaussian interpolation was introduced and evaluated using heatmap representations with Vik and Magma color scales (figure 8).

This incremental design process allowed isolating the perceptual effects of individual visualization parameters while maintaining a controlled comparison context. The following chapter evaluates these variants through an expert-based inspection, assessing each combination against the defined perceptual quality criteria.



Fig. 4: Bird’s-eye view of the VR environment showing a heatmap visualization using the Vik colormap with global normalization (fixed range: 10–40 °C).

## 5 Qualitative Evaluation of Visualizations

The evaluation was conducted as an expert-based inspection inspired by the approach proposed by Nielsen; Molich [NM90]. While the original method is based on general usability heuristics, this work employed a literature-based set of evaluation criteria specifically

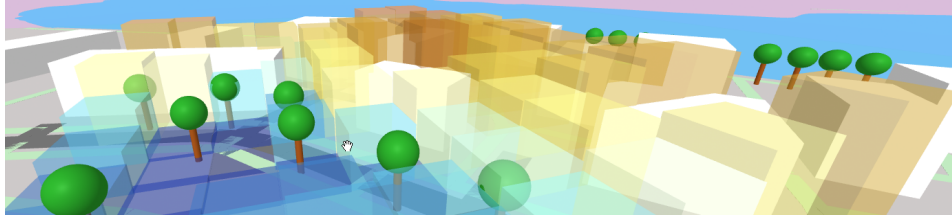


Fig. 5: Volumetric Heightmap Visualization of Temperature Data from a Bird's-Eye View

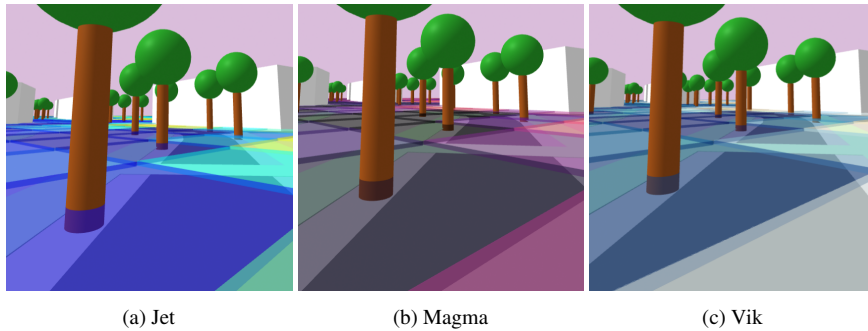


Fig. 6: Comparison of different color scales in a heatmap visualization from a first-person perspective.

developed to address the requirements of AR and VR visualizations. The objective of the analysis was to systematically assess the design alternatives developed in Chapter 4 (Figures 3, 4, 5, 6, 7, 8), which were generated from both a first-person perspective (FPP) and a bird's-eye view. The evaluation was carried out by a team of master's students involved in the project. To ensure an independent assessment, only team members who were not involved in the technical implementation of the prototype participated in the evaluation. All evaluators assessed the visualization variants independently and without prior discussion. Each evaluator rated the different visualization approaches according to the predefined criteria catalog. The individual ratings were subsequently aggregated by summing the assigned scores for each criterion and dividing the result by the number of evaluators. The resulting arithmetic mean served as the basis for comparing the different visualization approaches.

The evaluation considered different combinations of visualization type (heatmap and heightmap), color scale, and normalization method. These combinations were assessed using a set of predefined quality criteria:

- **Spatial Legibility (Hotspot Identification):** Spatial legibility describes how easily users can identify relevant spatial patterns and hotspots within the visualization. Since the primary objective of the proposed visualizations is to communicate urban heat distributions, the ability to quickly recognize areas of increased temperature is

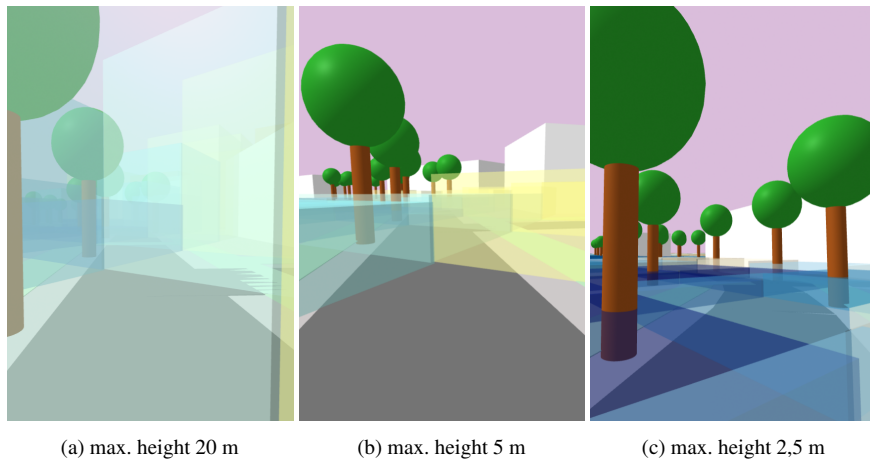


Fig. 7: Effect of height scaling parameters in the heightmap representation from a first-person perspective.

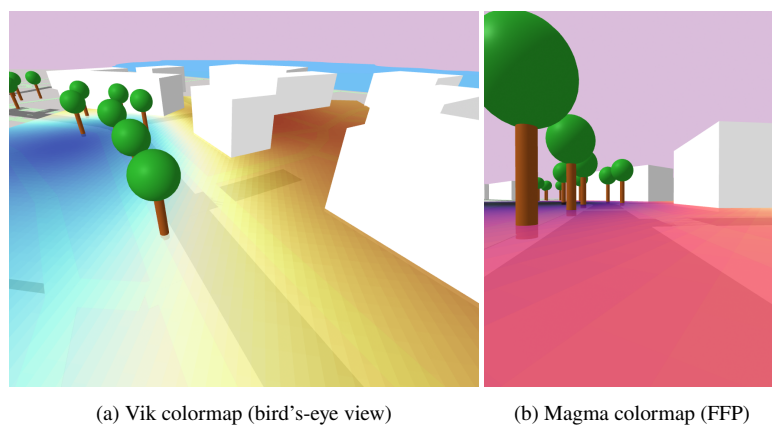


Fig. 8: Gaussian-interpolated temperature field showing continuous spatial distributions.

essential. This criterion is rooted in concepts of spatial legibility and visual perception [Ly64]; [Wa19].

- **Occlusion of Urban Geometry:** This criterion evaluates the extent to which the visualization obstructs existing urban structures and thereby interferes with spatial orientation. In situated AR applications, users must be able to simultaneously perceive both the virtual information and the physical environment. Excessive occlusion may reduce situational awareness and negatively affect usability [BFH01].
- **Color Discriminability:** Color discriminability measures how effectively different value ranges can be distinguished using the selected color scale. Since temperature information is primarily encoded through color, users must be able to reliably identify differences and gradients between areas. The criterion is based on established principles of cartographic color design and visual encoding [Br94].
- **Visual Coherence within the Urban Context:** Visual coherence assesses how well the visualization integrates into the surrounding urban environment without creating unnecessary visual distraction. As AR visualizations are superimposed onto real-world scenes, maintaining a balance between visibility and contextual integration is crucial for preserving the user's perception of the urban space [BCL15].
- **Data Interpretability:** Data interpretability describes how effectively the visual representation communicates the underlying temperature values and spatial relationships. Beyond merely highlighting hotspots, the visualization should support users in understanding the distribution and relative magnitude of temperature patterns. This criterion follows established principles of information visualization, which emphasize accurate and intuitive interpretation of data [Wa19].

The evaluation was conducted using a five-point rating scale ranging from 1 (*very poor*) to 5 (*very good*), where higher scores consistently indicate a more favorable assessment. Five-point Likert scales are widely used in evaluation studies, as they provide sufficient differentiation while supporting consistent and intuitive judgments by evaluators [Li32].

The evaluation primarily focused on the first-person perspective (FPP), as this represents the actual usage scenario of situated visualizations in augmented reality. However, a bird's-eye perspective was also included as a complementary viewpoint. While such a perspective is not typically used during on-site AR interaction, it provides valuable insights into spatial patterns and the overall distribution of temperature values. Furthermore, many AR applications incorporate overview or map-based views that allow users to inspect data from an elevated or abstract perspective. Evaluating both viewpoints therefore enables a more comprehensive assessment of the visualization approaches.

The evaluation revealed a trade-off between global comparability and local detail visibility. Global normalization facilitates direct comparison of temperature values across different areas, but tends to suppress local variations in regions with a limited temperature range. In contrast, local normalization enhances the visibility of local temperature differences and

Tab. 2: Expert ratings of the evaluated visualization variants (mean ratings; 1 = very poor, 5 = very good).

| Variant                     | Legibility<br>(FPP) | Legibility<br>(Bird's-eye) | Occlusion<br>(FPP) | Color Disc.<br>(FPP) | Color Disc.<br>(Bird's-eye) | Coherence<br>(FPP) | Data Interp.<br>(FPP) | Data Interp.<br>(Bird's-eye) | Total |
|-----------------------------|---------------------|----------------------------|--------------------|----------------------|-----------------------------|--------------------|-----------------------|------------------------------|-------|
| Heatmap<br>Jet              | 3.00                | 3.67                       | –                  | 3.67                 | 3.67                        | 1.33               | 3.33                  | 3.33                         | 22.00 |
| Heatmap<br>Magma            | 3.00                | 4.00                       | –                  | 3.00                 | 4.00                        | 4.67               | 3.33                  | 3.67                         | 25.67 |
| Heatmap<br>Vik              | 3.33                | 4.67                       | –                  | 4.00                 | 4.67                        | 4.33               | 3.67                  | 4.33                         | 29.00 |
| Heightmap<br>20 m           | 1.67                | 4.00                       | 1.00               | –                    | –                           | 1.33               | 2.33                  | 4.33                         | 14.67 |
| Heightmap<br>5 m            | 2.67                | 4.33                       | 3.33               | –                    | –                           | 3.33               | 4.00                  | 4.33                         | 22.00 |
| Heightmap<br>2.5 m          | 4.67                | 5.00                       | 4.67               | –                    | –                           | 4.67               | 4.67                  | 4.00                         | 27.67 |
| Raw Data                    | 3.67                | –                          | –                  | –                    | –                           | 2.67               | 4.33                  | –                            | 10.67 |
| Gaussian In-<br>terpolation | 4.00                | –                          | –                  | –                    | –                           | 4.67               | 3.67                  | –                            | 12.33 |

improves hotspot identification in the first-person perspective. However, this comes at the cost of reduced comparability between different spatial regions. For the intended situated visualization use case, local normalization was therefore considered the more suitable approach, as supporting on-site analysis and the identification of local temperature patterns was prioritized over cross-area comparison. The expert evaluation further indicated that the Vik heatmap achieved the highest overall rating (29.0/35), outperforming the Magma (25.67/35) and Jet (22.0/35) color scales. Although Jet provided comparatively good visual differentiation, it received the lowest ratings for visual coherence. The Vik color scale was rated highest for spatial legibility, color discriminability, visual coherence, and data interpretability, making it the preferred choice for first-person visualization.

For the heightmap visualization, a maximum height of 2.5,m achieved the highest overall score (27.67/30), providing the best compromise between spatial legibility, occlusion, visual coherence, and data interpretability. The 20,m variant suffered from severe occlusion and low visual coherence, whereas the 5,m variant represented an intermediate solution but remained clearly inferior to the 2.5,m configuration. Finally, the comparison of interpolation methods showed that Gaussian interpolation achieved a higher overall rating (12.33/15) than raw interpolation (10.67/15). While raw interpolation yielded slightly higher data interpretability ratings, Gaussian interpolation substantially improved visual coherence and spatial legibility, resulting in a more continuous and easier-to-interpret representation of the temperature distribution.

## **6 Discussion & Design Implications for Mobile AR**

### **6.1 Feasibility**

The prototype implementation demonstrates the technical feasibility of situated visualizations of urban heat data. The results show that geospatial data based on a  $10 \times 10$  m grid can be successfully processed, provided through an API, and spatially anchored within a WebVR environment. This enables abstract climate data to be experienced directly within its real-world spatial context, thereby creating new opportunities for communicating urban environmental risks. Overall, the findings support the assumption that immersive technologies can facilitate a more intuitive and accessible understanding of complex environmental information.

### **6.2 Design Trade-offs**

The development process revealed several important design trade-offs. One major trade-off concerns the balance between comparability and local detail in color scaling. The dataset-dependent color scale used in this work increases the visibility of local temperature variations but may also lead to misleading interpretations. For example, relatively cool areas on particularly hot days may still be displayed using colors that are commonly perceived as "cold," despite representing comparatively high temperatures. In contrast, a fixed color scale would improve comparability across different datasets but could reduce the visibility of subtle local differences. A second trade-off exists between information density and spatial orientation. Volumetric visualizations, such as height maps or heat clouds, effectively communicate temperature variations but partially occlude the real environment, thereby reducing users' spatial awareness. For mobile AR applications, ground-based visualizations therefore appear more appropriate, as they preserve the connection to the physical environment while maintaining visibility of surrounding objects. Another challenge lies in balancing data accuracy and visual coherence. Interpolation techniques such as Gaussian filtering help reduce abrupt transitions between raster cells, producing smoother and more visually coherent representations. However, the resulting visualization deviates from the originally measured values, meaning that increased visual realism comes at the cost of reduced data fidelity. This trade-off should be carefully considered in future visualization designs. An additional limitation concerns the seasonal validity of the underlying dataset. The climate simulations used in this study represent a typical summer day. Consequently, the results cannot easily be generalized to other seasons, since microclimatic processes vary considerably throughout the year. The presented heat patterns therefore provide only limited temporal validity.

### 6.3 Technical Implications for Mobile AR

Transferring the developed VR prototype to mobile AR environments introduces several additional technical challenges. Unlike VR, AR applications must continuously and accurately align virtual content with the physical environment. This requires robust tracking methods and real-time processing of geospatial data. Furthermore, large raster datasets impose substantial scalability requirements, making efficient data reduction and streaming techniques essential. A promising approach is server-side hosting of geospatial datasets combined with dynamic loading of only those spatial regions required for the user's current location. Closely related to these challenges is energy consumption. Mobile AR applications require continuous sensor processing, real-time tracking, and rendering, making them computationally demanding and increasing battery usage on mobile devices [Ga16]. Consequently, applications designed to raise awareness of climate-related issues should also consider their own environmental footprint. Energy-efficient software engineering ("green coding") therefore becomes an important design objective, for example through adaptive data streaming, efficient tracking algorithms, and energy-aware rendering [Ju23]. This dilemma highlights the importance of energy-efficient design strategies for future mobile AR systems.

### 6.4 Limitations

Several limitations should be considered when interpreting the findings of this study. First, no formal user study was conducted. Consequently, no conclusions can be drawn regarding the usability, comprehensibility, or potential influence of the visualizations on users' perception of heat-related risks. Second, data reduction was achieved solely through a static crop of a  $100 \times 100$  pixel region, corresponding to approximately one square kilometer. Dynamic adaptation of the displayed area based on user location or interaction was not implemented. Third, the investigation focused exclusively on the Tunnelstraße area in Berlin. As a result, the transferability of the findings to other urban environments or larger study areas remains uncertain. Overall, the results demonstrate that situated visualization of urban heat data is technically feasible and holds considerable potential for communicating climate-related environmental risks. At the same time, the identified design trade-offs and technical challenges indicate that additional research and development are required before such visualizations can be effectively deployed in practical mobile AR applications. ““

## 7 Conclusion and Future Work

This thesis investigated the potential of Augmented Reality (AR) for situated visualization of urban heat exposure using Berlin as a case study. The results demonstrate that geospatial temperature data can be integrated into immersive environments, enabling abstract environmental information to be experienced directly within its spatial context. Three main

contributions were achieved. First, a reproducible pipeline was developed for processing and integrating geospatial raster data using established GIS methods, providing a reusable basis for future environmental visualization projects. Second, an open WebVR prototype was implemented, offering a flexible platform for evaluating different visualization approaches and serving as a technical foundation for future mobile AR applications. Third, a systematic comparison of visualization techniques identified a ground-based heatmap combined with the perceptually uniform diverging color map *Vik* and local normalization as the most suitable approach for communicating urban heat patterns while preserving spatial orientation.

The study also identified several challenges related to data resolution, visualization design, and technical implementation. In particular, mobile AR applications require accurate tracking, scalable geospatial data management, and computationally efficient rendering to ensure usability on resource-constrained devices.

Future work should focus on formal user studies to evaluate usability, risk perception, and the effectiveness of situated environmental visualizations. Furthermore, the prototype should be transferred to a fully functional mobile AR application to investigate tracking accuracy, interaction techniques, and real-world performance. Finally, scalable server architectures for dynamic geospatial data streaming and the extension of the proposed approach to additional environmental phenomena, such as air pollution or noise exposure, represent promising directions for future research. Overall, this work demonstrates that AR is a promising medium for communicating urban environmental risks. However, realizing its full potential requires further advances in user-centered design, scalable geospatial infrastructures, and energy-efficient mobile AR systems.

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